

Multiple Choice (10 marks)

1. Obtain the partial-fraction expansion of $F(s) = [(s + 2) / \{(s + 1)^2 (s + 3)\}]$
 - (a) $F(s) = [(1 / 5) / (s + 1)] + [(2 / 5) / (s + 1)^2] - [(2 / 5) / (s + 3)]$
 - (b) $F(s) = [(1 / 2) / (s + 1)] - [(1 / 2) / (s + 1)^2] + [(1 / 4) / (s + 3)]$
 - (c) $F(s) = [(3 / 4) / (s + 1)] - [(1 / 8) / (s + 1)^2] - [(5 / 8) / (s + 3)]$
 - (d) $F(s) = [(1 / 2) / (s + 1)^2] + [(1 / 4) / (s + 1)] - [(1 / 4) / (s + 3)]$
 - (e) $F(s) = [(1 / 6) / (s + 1)] + [(1 / 3) / (s + 1)^2] - [(1 / 2) / (s + 3)]$
2. The smallest change in measured variable to which instrument will respond is
 - (a) calibration.
 - (b) sensitivity.
 - (c) resolution.
 - (d) repeatability.
 - (e) linearity.
3. Use Laplace's equation to find the capacitance of a parallel plate capacitor.
 - (a) $C = (\epsilon^2 S / d)$
 - (b) $C = (/ d^2)$
 - (c) $C = (/ d)$
 - (d) $C = (\epsilon / dS)$
4. The bandwidth and cut-off frequency of an antenna are 0.8 MHz and 44 MHz respectively. Calculate the Q-factor of the antenna.
 - (a) 55
 - (b) 50
 - (c) 75
 - (d) 85
 - (e) 65
5. Compute the impact stress in a steel rod which is hit by a block weighing 45 lbs. The speed of the block is 40 in./sec. The rod is 18 in. long and has a diameter of $1(1 / 4)$ in.
 - (a) 8,762.3 psi
 - (b) 4,321.0 psi

(c) 25,678.9 psi

(d) 6.25 lbs

(e) 30,000 psi

True / False (10 marks)

Answer True or False

6. True or False: The one cycle surge current for an SCR rated at 3000 A for half cycle is approximately 3000 A.
7. True or False: Inserting a dielectric slab into a parallel plate capacitor decreases the stored energy.
8. True or False: The heat transferred in this scenario is 100J, indicating energy is absorbed by the system.
9. True or False: The probability $\Pr[0 < x < (3/4), (1/3) < y < 2]$ equals $1/4$, which is a common error.
10. True or False: The entropy change can be expressed as $ds = C_v (dT/T) + (R/J)(dp/p)$ for ideal gases.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the thermodynamic processes involved when saturated steam at 100 psia from a steam pipe enters an empty cylinder. What will be the final state of the steam and why?
12. Discuss the probability of obtaining 2 heads and 1 tail when tossing a penny three times. Analyze the underlying principles of probability that apply to this scenario.

End of Paper